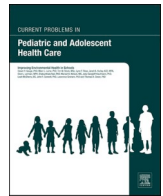




Contents lists available at ScienceDirect

Current Problems in Pediatric and Adolescent Health Care

journal homepage: www.elsevier.com/locate/cppeds

Competency based medical education – Where do I start?

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ARTICLE INFO

Keywords:

Competency based medical education

Entrustable professional activities

Competencies

Milestones

Entrustment

Unsupervised practice

Introduction

Competency-based medical education (CBME), rooted in the broad fundamentals of outcomes-oriented education and training, includes the demonstration of behavior-based role-derived competencies with mastery levels and accountability to the public. A key principle of CBME, the outcomes principle,¹ holds the idea that accountability to the public and to patients as critical, and incorporates caring for patients as the ultimate purpose of medical training. Thus, CBME provides a framework for lifelong learning and ongoing assessment designed to ensure that physicians are prepared to meet the health care needs of the communities they serve.

Competency frameworks generally describe the qualities or attributes that a professional, such as a doctor, should acquire and serve as the foundational core component of CBME.² An overall competency

framework forms a theoretical construct in need of translation into a curriculum and program of assessment by those responsible for educating learners.

The foundation of a competency framework in medicine is defining the needs of the patients and using these needs to determine curricula and a program of assessment.

This program of assessment should include not only the competencies and developmental milestones of individuals as they progress through a training program but should also include the activities needed by patients.

Abbreviations: ACGME, Accreditation Council for Graduate Medical Education; ABMS, American Board of Medical Specialties; ABP, American Board of Pediatrics; CCC, Clinical Competency Committee; CBD, Competence by Design; CBME, Competency Based Medical Education; EPAs, Entrustable Professional Activities; GME, Graduate Medical Education; LOS, Level of Supervision; PCs, Program Coordinators; RMS, Resident Management System; STAR, Statement of Awarded Responsibility.

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<https://doi.org/10.1016/j.cppeds.2024.101674>

Available online 31 August 2024

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Entrustable professional activities (EPAs), which are units of professional practice, represent the tasks and responsibilities that physicians do to provide care to patients and define the activities expected in the profession.

Our specialty has developed an EPA framework for general pediatrics and pediatric subspecialties to serve as the foundation for education and assessment in our training programs.³ Each of these EPAs integrate multiple competencies, and for an individual to successfully execute an activity, they must demonstrate a number of specific behaviors related to a given activity, reinforcing the importance of a comprehensive program of assessment that includes readiness to execute an activity *and* the developmental behaviors accomplished along the way.

However, while consideration of the behaviors and specific competencies required to perform an activity are an important part of a program of assessment, care is needed to avoid distillation of competency into check lists of activities resulting in the conceptual state of a person being mistaken for their performance behavior.⁴ The concept of entrustment serves to address this issue by considering a trainee's readiness to execute an EPA in a holistic manner.

Entrustment decisions granting full trust to a trainee to do an EPA without supervision should be rooted in the trainee's achievement of all the competencies needed to successfully complete the activity. In thinking of entrustment, there are four important variables to consider: 1) attributes of the trainee, 2) attributes of the supervisor, 3) the context of the activity, and 4) the nature of the EPA (i.e., simple/complex or frequent/common).⁵ Additionally, entrustment decisions may be made in the moment during, or just before, a clinical encounter or made more structurally by the Clinical Competency Committee (CCC) in a summative fashion once the trainee has passed a level of supervision (LOS) threshold resulting in a Statement of Awarded Responsibility (STAR).⁵

When the concept of entrustment is integrated into the current program of assessment in pediatrics, data demonstrate substantial gaps in trainees' readiness to execute EPAs at the time of graduation.⁶⁻⁸ These data reinforce the importance of developing an authentic approach to CBME that focuses on trainees' ability to achieve the EPAs that have been defined by our community as essential for all pediatricians and pediatric subspecialists. In this article, we aim to provide information on the history of CBME, the framework and core components of CBME, suggestions on how to begin implementation of CBME system, and experiences with implementation of CBME. Through a pragmatic approach, this article follows a concise outline aimed to assist GME leaders, faculty, and trainees as they embark upon the complex yet rewarding journey of authentically implementing CBME.

History of competency based medical education

Competency based education and training is a concept that is now over 50-years-old.^{1,9-13} It importantly includes the demonstration of behavior-based role-derived competencies, assessments with mastery levels, the idea that progress be measured based on demonstration of competency, and accountability to the public.^{1,9-13} In 1978, McGahie, et al. published a report to the World Health Organization in which the CBME model was promoted as optimal medical education, as opposed to traditional "subject-centered" curricula.¹⁴ Despite these recommendations, broad and genuine implementation has not yet been achieved.

However, over the last two decades, efforts to implement CBME as the expectation of medical training and assessment worldwide are gaining momentum. In the United States, in 2001, the ACGME and the American Board of Medical Specialties (ABMS) jointly supported the Outcomes Project, which represents a first step from a process-based educational framework to one built on the principles of CBME.^{15,16}

Additional steps were taken toward a more competency-based model in 2012 when the ACGME transitioned to a revised accreditation system.¹⁷ The next steps in our journey to achieving full implementation of CBME is a *focus on outcomes*, as described by Frank, et al. as a foundational modern core principle of CBME.¹ To that end, the American Board of Pediatrics has committed to including our specialty's outcomes, the EPAs, as part of certification decision-making by 2028. The *focus on outcomes* principle¹ is rooted in the idea that accountability to the public and to patients is critical, and medical educators have the responsibility of ensuring to the public that medical graduates are competent in their intended practice. This requires that, at minimum, outcomes be explicitly defined, curricula designed intentionally to address these outcomes, and trainees be assessed accordingly for their progressive achievement of competence toward the desired outcomes.

These outcomes, which are defined by the specialty as essential abilities for providing care to that specialty's patient population, are represented by a set of EPAs, and assessment of these activities is currently missing from many pediatric training programs in the United States.¹⁸

The other three modern CBME principles defined by Frank et al.¹ (*emphasis on abilities*, *de-emphasis on time-based training*, and *trainee-centeredness*) together reflect the key components of the educational process to ensure each trainee achieves these desired educational outcomes through tailored learning experiences shaped by the trainee, yet while based on patient needs. In this way, educational (or trainee) outcomes and desired patient outcomes are aligned and integrated. The transition to CBME also provides the opportunity for refreshed conversations related to design, redesign, and implementation of curricula and also the role of assessment practices in facilitating the path to competence and readiness for practice for individual trainees.

The framework and core components for designing a system of competency based medical education

CBME represents a framework for lifelong learning and ongoing assessment, which is an important priority for physicians given that medicine is a self-regulating profession. A robust CBME framework should be designed to ensure that physicians are prepared to meet the health care needs of the communities they serve and includes five core components: 1) outcomes-based competencies, 2) progressive sequencing, 3) tailored learning experiences, 4) competency-focused instruction, and 5) programmatic assessment.² Integrating these 5 core components into the entire continuum from training through practice is essential for a comprehensive approach to ongoing learning and assessment to optimize patient care.

The foundation of any CBME model is the first component, defining the outcomes expected in a given specialty. The pediatrics community, in partnership with the ABP, defined EPAs for general pediatrics¹⁹ and the pediatric subspecialties²⁰ almost a decade ago. EPAs which can be thought of simply as 'everyday pediatrician activities', help to define the desired educational outcomes for our specialty, and are the routine, observable units of professional practice or activities that patients can expect from their pediatricians.³ When EPAs are combined with ACGME competencies and Milestones, a comprehensive outcomes-based framework has been created and can be used throughout the entire continuum for education and assessment. The EPAs represent an important addition to the traditional competency and Milestone based approach to assessment in US GME training programs given that these activities bring patient needs to the forefront in education and assessment.

As these outcomes are considered, the second core component,

progressive sequencing of competencies becomes an important consideration. CBME prioritizes a trainee's progression and acknowledges that learning does not necessarily follow a linear sequence. Learning experiences in the training environment should provide trainees with experiences that resemble the practice environment. This leads to our third core component, tailored learning experiences. Trainees enter into GME with different strengths and growth opportunities. Further, we should expect trainees to demonstrate mastery of competencies at different stages of their professional development. The EPAs can be used to guide the learning process by also informing individualized rotational goals and objectives to provide a foundational base while allowing trainees to grow their knowledge base, skills, and attitudes at their own pace until achieving readiness for unsupervised practice for the various EPAs. While the ability to execute the EPAs are likely closely related, most EPAs do not require achievement of one in order to execute another, which allows for substantial flexibility in how they are organized in the curriculum at the program level or even for individual trainees. As programs implement an outcomes-based framework, faculty and trainee development will be important for the fourth core component, competency focused instruction, whereby involved individuals work collaboratively understanding their respective roles in the mentor/coach-trainee relationship and in developing or refining individualized learning plans based on the defined educational outcomes.

The fifth and final core component of a CBME model is programmatic assessment. Stated differently, this is a comprehensive portfolio-based approach to assessment of a trainee's competence that includes multiple data points to facilitate decision-making about progression of the trainee towards readiness for unsupervised practice as a pediatrician or pediatric subspecialist.

A CBME model requires assessment of trainees that occurs often, preferably through real-time authentic observation as the trainee provides care for patients, by multiple assessors, and integrates the data into constructive behaviorally based feedback and coaching for the trainee.

Frontline faculty will require training to equip them with the tools and resources to make accurate assessments and provide quality verbal and written feedback to trainees that is timely, specific, action-oriented, and developed in partnership with the trainee with the goal of achieving competence. These assessments can be collated into a comprehensive and well-rounded portfolio that can guide a trainee's progression and aid the Clinical Competency Committee (CCC) in making progression, advancement, and entrustment decisions. This method allows programs to identify trainees that may be struggling earlier in training so that their education can be tailored to help them achieve the necessary goals by the completion of training. The data can also be used by programs to ensure that their approach to assessment is equitable and to identify areas of strength and opportunities for improvement at the program level.

Suggestions on how to begin implementation of competency based medical education

To recap, a fully realized CBME system with the 5 core components requires defined educational outcomes, such as EPAs, with progressive sequencing; a flexible curriculum that facilitates achievement of these outcomes through tailoring based on an individual trainee's progress; educators who support trainees in reaching competency-based outcomes; and robust assessment systems incorporating timely and frequent feedback that will accurately capture trainee progress towards those outcomes.^{2,21}

Transitioning to a genuine CBME system can be a daunting endeavor, requiring a dynamic growth mindset, and it is important to consider the multiple layers associated with implementation!

Given the importance of local context and differences in program size, setting, and structure, there is not a simple and singular approach that applies to all circumstances. Some guidance regarding implementation on a national scale can be learned from the Canadian experience, as the Royal College of Physicians and Surgeons has recently implemented their "Competence by Design (CBD)" program across all medical specialties in Canada.²² An early step in the complex endeavor of transitioning to a CBME system is setting your program up to address defined educational outcomes. We suggest the following steps to help with this aspect while appreciating that all the components of a robust complete CBME system are interlinked.

1. Assess your program's readiness for change. Determine the general level of understanding about CBME, existing facilitators and barriers to the transition, and the resource needs for building readiness for the process.²²
2. Build a team through identification of stakeholders such as departmental leaders, faculty, trainees, program coordinators (PCs), and others who will be impacted by the transition to CBME.²²
3. Familiarize yourself with the names, descriptions, and context for the appropriate pediatric¹⁹ or pediatric subspecialty EPAs²⁰. EPA language will need to be thoughtfully integrated into your program's curriculum, rotation goals and objectives, and various assessment forms.
4. Map the EPAs to each rotation or experience in your curriculum (Fig. 1). Curriculum mapping helps you to:
 - a. Determine which EPAs are best assessed in each curricular experience.
 - b. Reveal gaps in your curriculum and identify any EPAs that are not adequately represented.
 - c. Plan learning activities and educational strategies, based on the included EPAs as educational outcomes, that can be assessed in specific curricular experiences.
 - d. Limit the scope of a given individual faculty member in terms of which EPAs they should be most familiar.
5. Develop assessments to assess the EPAs that have been mapped to your curriculum.
 - a. Quality assessments typically are brief, taking only a few minutes to complete, are user friendly with clear and concise instructions and rating scales/anchors, address an important clinical activity, i.e., EPA, and ensure that there is opportunity for narrative comments with concrete and specific actionable feedback individualized to the trainee.^{23,24} Note that faculty in most rotations will only be responsible for assessing trainees on a few EPAs, allowing them to develop expertise in a specific EPA. Many faculty report that assessment of entrustment using EPAs is more straightforward than traditional milestone-based assessment.²⁵
 - b. Modify your program's Resident Management System (RMS) to include EPAs based on your work outlined in the steps above. The RMS will still need to generate Milestone reports for submission to the ACGME and to guide trainee development. On that note, the ABP has partnered with the pediatrics community to judiciously map the ACGME Milestone subcompetencies to each EPA. Subcompetency milestone ratings-EPA LOS mapping allows programs to efficiently generate data on both Milestones and EPAs while limiting the assessment burden on faculty and the CCC. In general pediatrics, a process will be available soon to generate suggested

EPA	Description	Continuity Clinic Faculty	Pediatric Hospital Medicine Faculty	Inpatient Subspecialty Faculty	Pediatric Emergency Medicine Faculty	Newborn Faculty
EPA 1	Provide consultation to other health care providers caring for children					
EPA 2	Provide recommended pediatric health screening	X				X
EPA 3	Care for the well newborn					X
EPA 4	Manage patients with acute, common diagnoses in an ambulatory, emergency, or inpatient setting	X	X	X	X	
EPA 5	Provide a medical home for well children of all ages	X				
EPA 6	Provide a medical home for patients with complex, chronic, or special health care needs	X	X	X		
EPA 7	Recognize, provide initial management and refer patients presenting with surgical problems				X	
EPA 8	Facilitate the transition from pediatric to adult health care	X				
EPA 9	Assess and manage patients with common behavior/mental health problems	X	X			
EPA 10	Resuscitate, initiate stabilization of the patient and then triage to align care with severity of illness		X	X	X	
EPA 11	Manage information from a variety of sources for both learning and application to patient care	X	X	X	X	X
EPA 12	Refer patients who require consultation		X	X	X	X
EPA 13	Contribute to the fiscally sound, equitable and collaborative management of a healthcare	X				
EPA 14	Apply public health principles and quality improvement methods to improve population					
EPA 15	Lead an interprofessional health care team		X	X		
EPA 16	Facilitate handovers to another health care provider either within or across settings		X	X	X	
EPA 17	Demonstrate competence in performing the common procedures of the general pediatrician					

Fig. 1. EPA-Assessment Mapping 2024. Image credit, Dr. David A. Stewart.

milestone ratings from EPA assessments based on a multi-year, multi-center study that is almost complete. In the subspecialties, a working group in each subspecialty has developed specific mapping based on expert consensus that will also allow for automated generation of suggested milestone ratings. Programs may choose to use these milestone rating-EPA LOS mappings, or generate their own, to help the CCC extrapolate Milestone data from EPA data.²⁶

- c. Devise an approach for distribution of assessments. Ideally, these assessments should be integrated into the clinical workday and available as close to a clinical encounter as possible (i.e., at the end of a clinic day or after seeing an inpatient consultation). A process to prompt assessments by faculty, trainees, or administrators is opportune. Program Coordinators will be essential partners; assessments can then easily be collated by PCs, in preparation for the CCC especially, through use of your RMS.

6. Create a CCC who will be responsible for interpreting multiple sources of assessment data and making advancement, progression, and entrustment decisions based on *competency* for each trainee.²³
7. Provide trainee and faculty development to educate stakeholders on CBME, EPAs, and the benefits to the patients and families.²⁷

Experiences with implementation of competency based medical education

There is substantial time and effort involved in making the changes needed to do CBME well; dedicated time for program leaders and buy-in from all stakeholders are essential to success.^{28,29} As we each locally work to develop a more competency-based approach to education and assessment in GME, there are many resources and partners to consider, including the Royal College of Physicians and Surgeons of Canada²², education leaders from a variety of other countries, and many other

specialties in the United States.^{22,30} Within pediatrics, we can learn from a pediatric pulmonology fellowship program who recently described their CBME journey.³¹ Leaders in this program felt timing was right to overhaul the program to integrate the principles of CBME and EPAs due to repeated recruitment difficulties and faculty disengagement. To address major program deficiencies, they implemented an EPA-based curriculum, mapped the associated EPAs to ACGME Milestones, and provided faculty with a less burdensome trainee assessment form. Faculty surveyed afterwards preferred the ease of a holistic assessment which allowed room for comments and anecdotes, compared to the previous Milestone-based assessment. In addition, the implemented changes led to a better understanding of their trainee's achievement of competency during CCC meetings, as well as which Milestone deficiencies needed to be addressed, allowing for a direct and focused trainee performance improvement plan.^{26,31} Nationally, the ACGME-ABMS CBME Learning Community and ABMS CBME Interest Group are guiding the transformation of medical education to CBME broadly across specialties in the US.^{32,33} Other boards, such as the American Board of Surgery, the American Board of Family Medicine, and the American Board of Orthopedic Surgery have implemented a competency-based approach to training and certification, plus many other specialties, including Pathology, Emergency Medicine, Genetics/Genomics, and others are at various stages in their journey toward a competency-based system.³³

Summary

By discussing the history of CBME along with the framework and core components for a system of CBME, suggestions on how to begin, and shared experiences, we have offered a short course on CBME lending to your improved understanding and anticipate your empowerment to embark upon an implementation journey. Although CBME is not new, we do appreciate the approach to medical education in this regard, including the use of EPAs, may be quite new to pediatric programs in a broad sense. As the needs of pediatric patients change over time, curricula, as well as EPAs, will need to be revisited, adapted, or developed to ensure trainees and practicing pediatricians are achieving the desired outcomes for our specialty. Program evaluation and research will be informative to furthering our understanding on the critical elements for trainee development and achievement of competency and for faculty accuracy and confidence in making entrustment decisions, and if CBME as a system for GME results in better trainee and patient outcomes.

Disclosures

Ms. Nuncio Lujano and Dr. Turner are employed by the American Board of Pediatrics. The content of this manuscript is solely the responsibility of the authors and does not necessarily represent the official views of the American Board of Pediatrics or the American Board of Pediatrics Foundation.

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